

2016.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

COOLING SYSTEM DRAINING, FILLING AND BLEEDING (G1514977)

GENERAL PROCEDURES

DRAINING

CAUTION:

Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.

NOTE:

This procedure only needs to be carried out when the coolant system vacuum filler equipment is not available in the workshop.

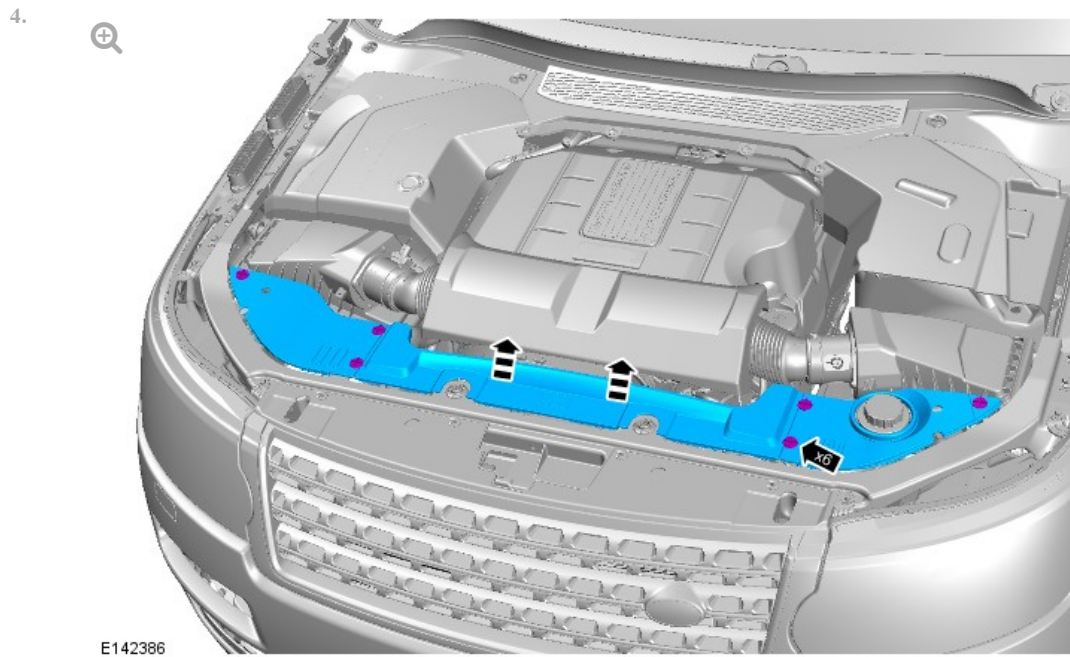
1.

WARNING:

Make sure to support the vehicle with axle stands.

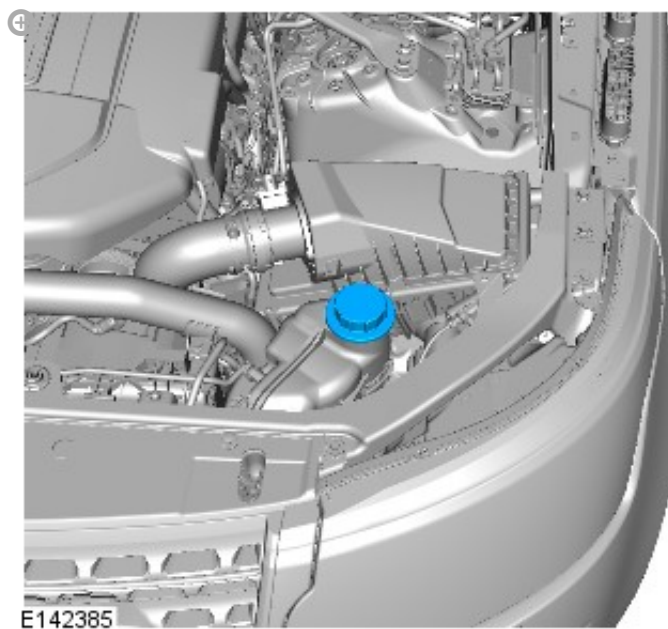
Raise and support the vehicle.

2. Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).
3. Refer to: [Engine Cover - TDV6 3.0L Diesel](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).



5. **WARNING:**

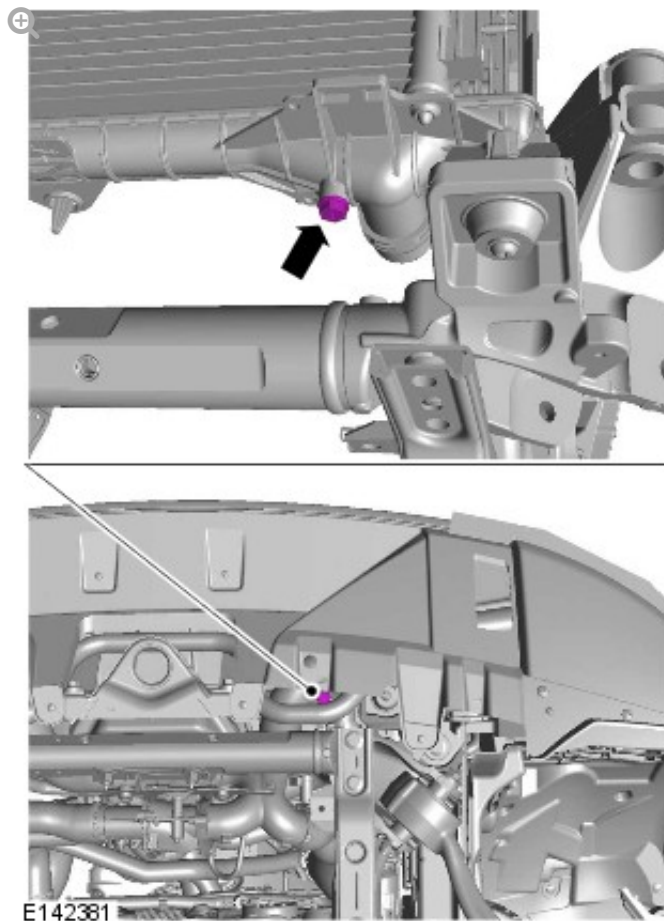
When releasing the cooling system pressure, cover the coolant expansion tank cap with a thick cloth.



6.

CAUTION:

Be prepared to collect escaping coolant.



NOTE:

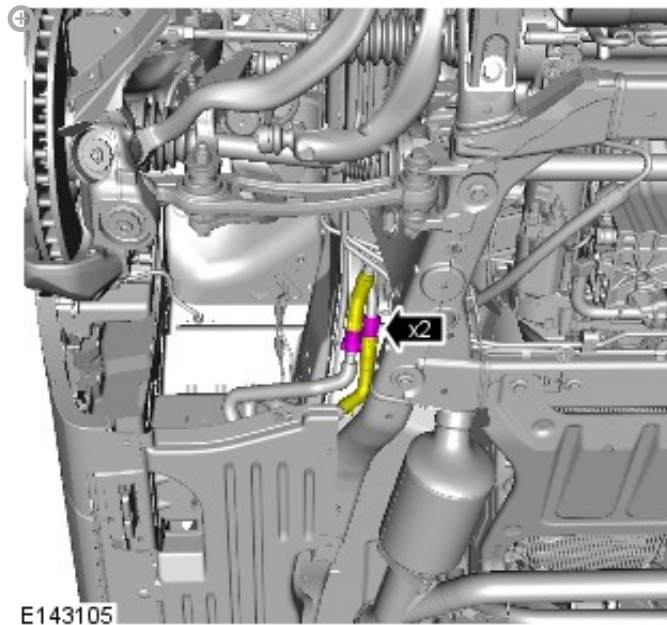
Step 7-10 and 12-13 only needs to be carried out when auxiliary climate control system has been fitted to the vehicle.

7. Remove the front RH wheel arch liner.
Refer to: [Fender Splash Shield](#) (501-02 Front End Body Panels, Removal and Installation).

8.

CAUTION:

Be prepared to collect escaping coolant.

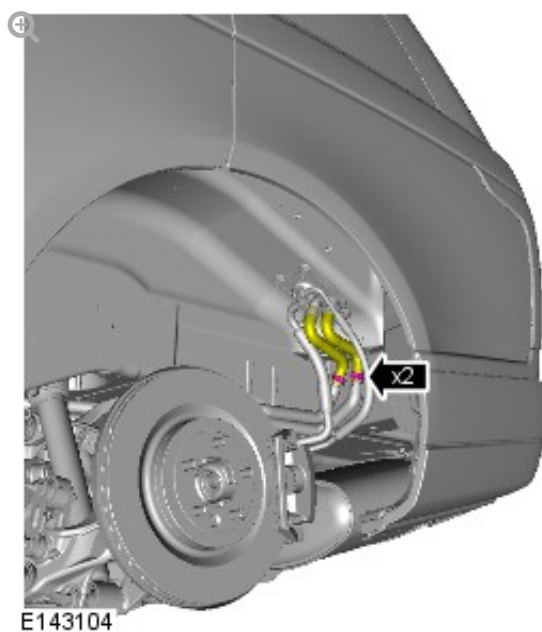


9. Remove the rear LH wheel arch liner.
Refer to: [Rear Wheel Arch Liner](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

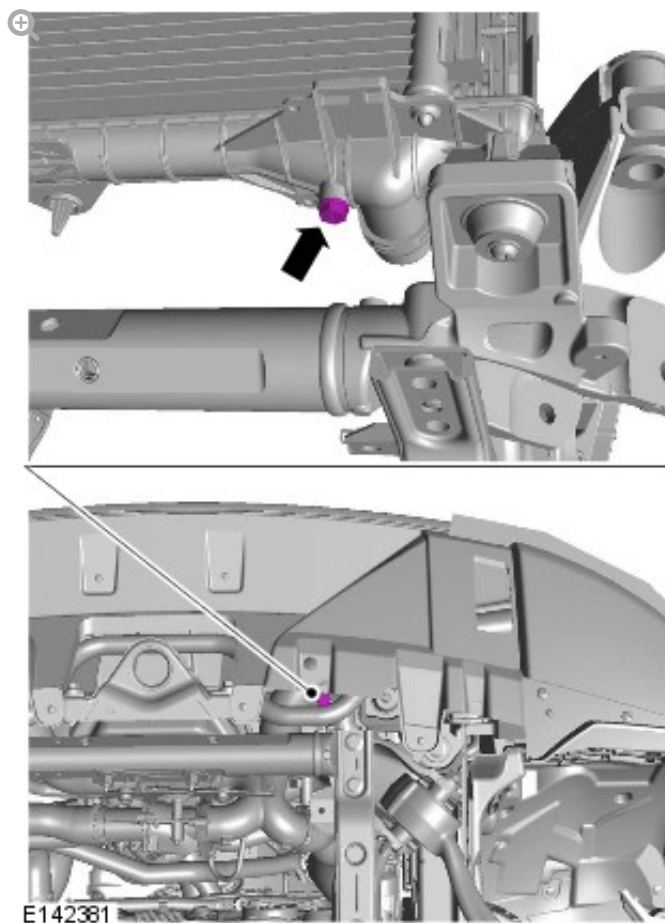
10.

CAUTION:

Be prepared to collect escaping coolant.

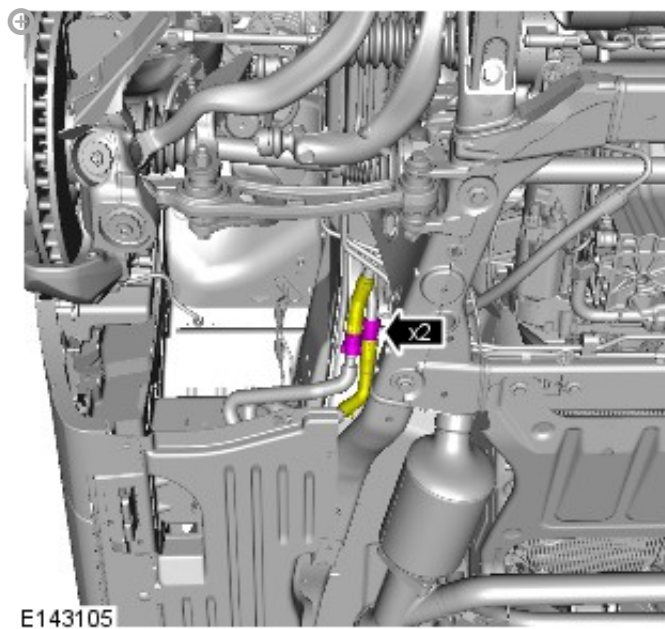


11.

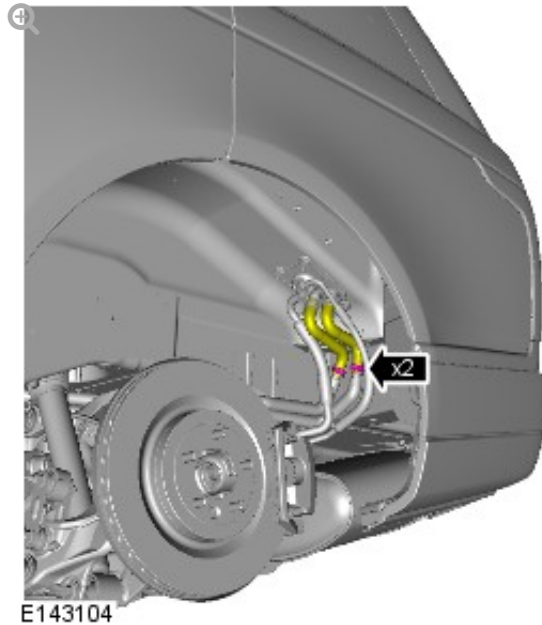


Torque: 5 Nm

12.



13.



14. Carry out the procedure up to step 1 in the filling section three times, filling the cooling system with clean water after the first two drains. At the third refill, use a suitable measuring tool to make sure that the cooling system maintains a 50% mix.

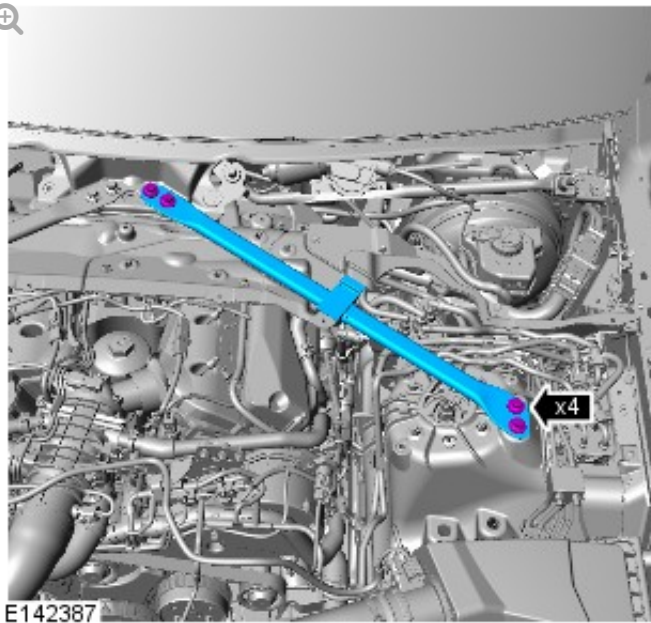
NOTE:

Step 7-10 and 12-13 only needs to be carried out when auxiliary climate control system has been fitted to the vehicle.

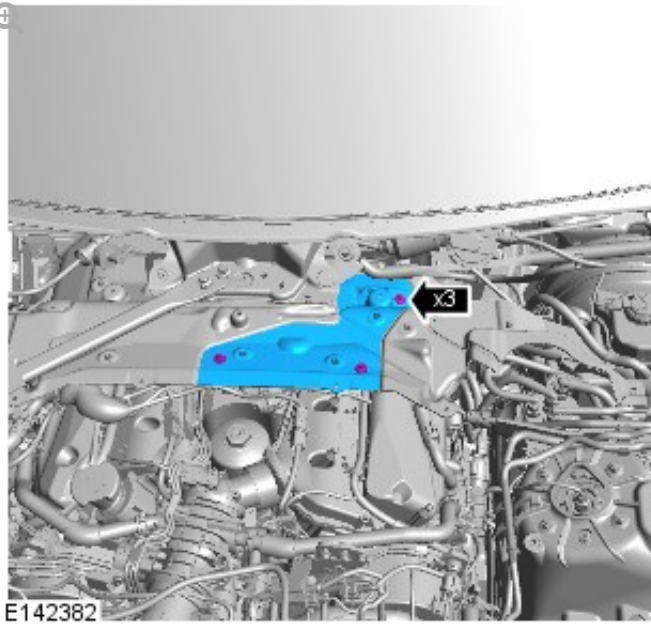
FILLING

15. Refer to: [Cowl Panel Grille](#) (501-02 Front End Body Panels, Removal and Installation).

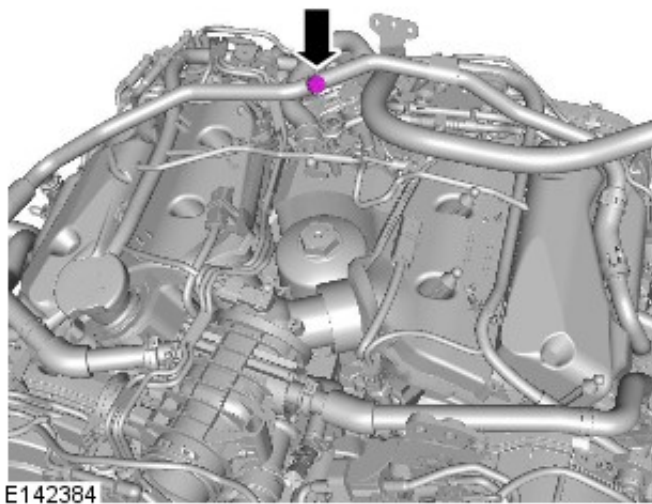
16.



17.

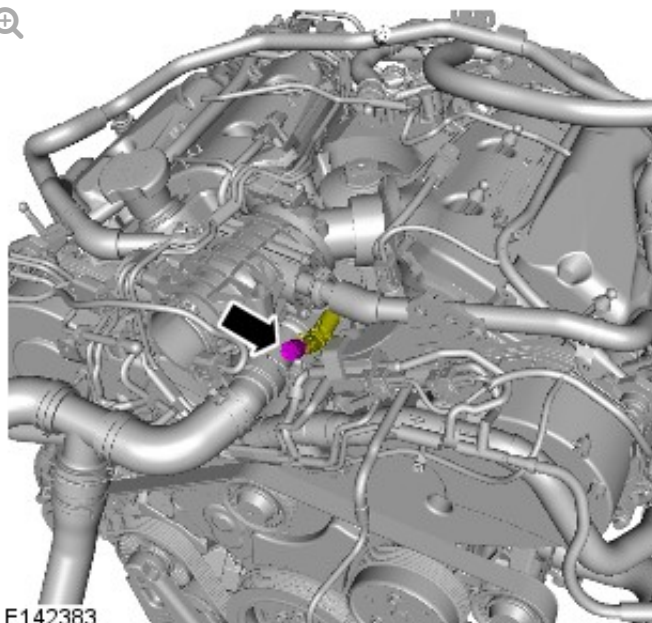


18.



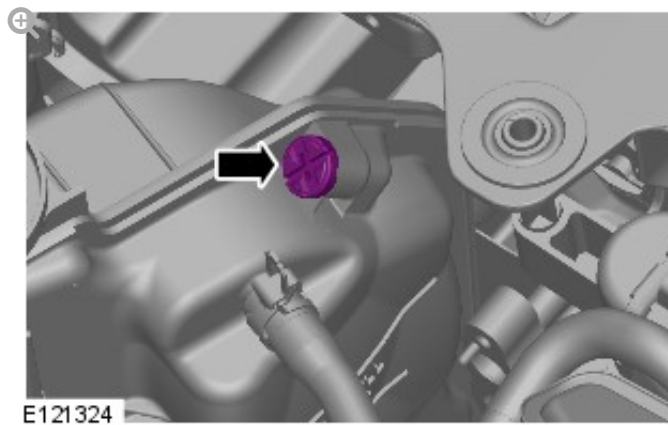
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19.



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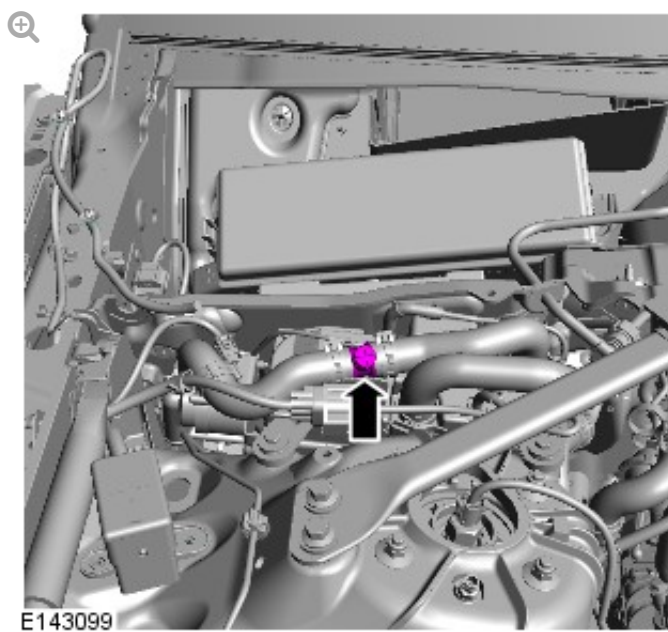
20.



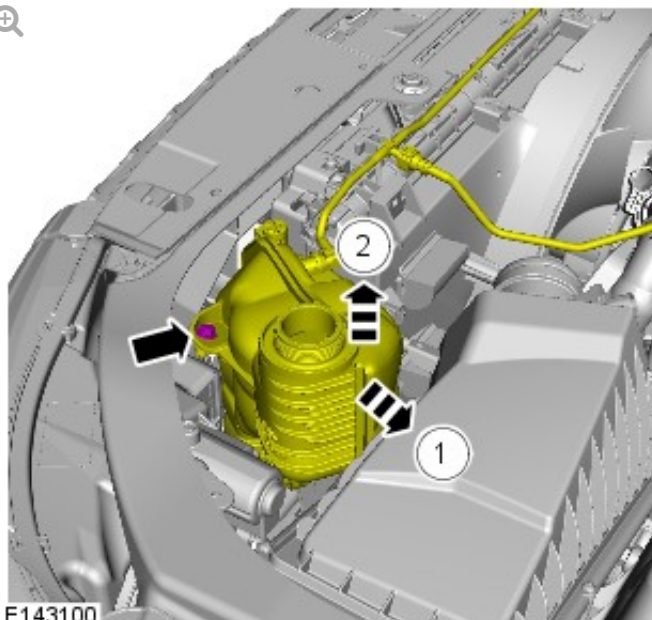
21.

NOTE:

Where fitted.



22.

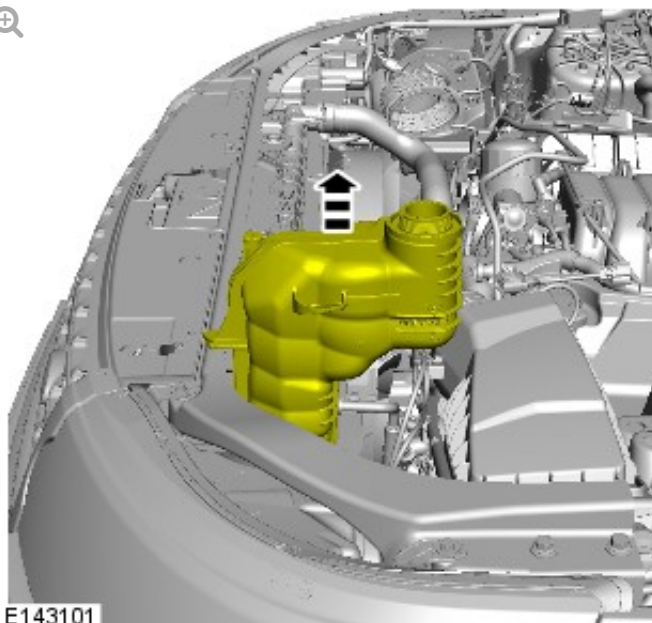


Torque: 9 Nm

23.

NOTE:

Keep the coolant expansion tank vertically.



Raise the coolant expansion tank minimum 110mm from its original position.

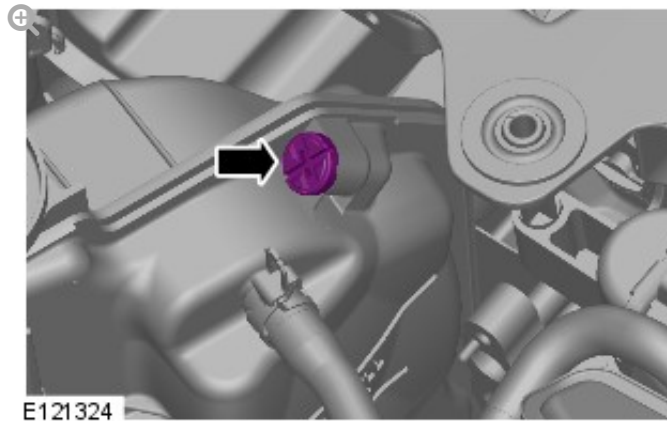
24.

CAUTION:

Anti-freeze concentration must be maintained at 50%.

Fill the coolant expansion tank to the rim and continue until coolant is visible from the outlet.

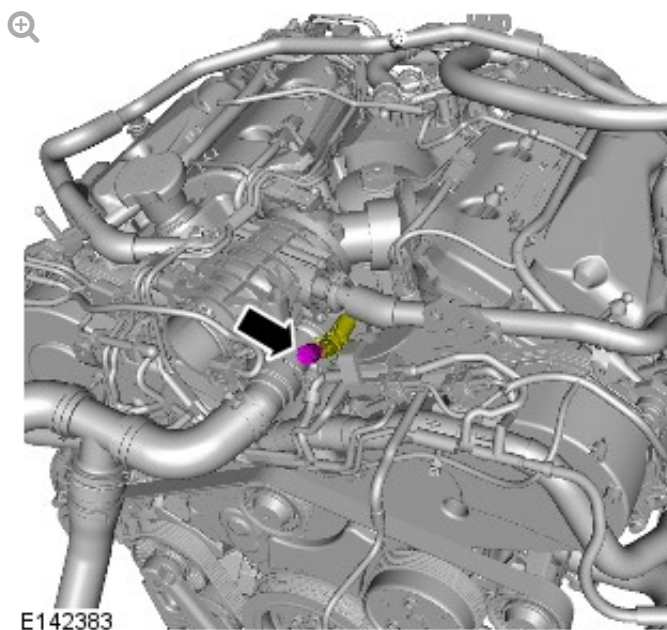
25.



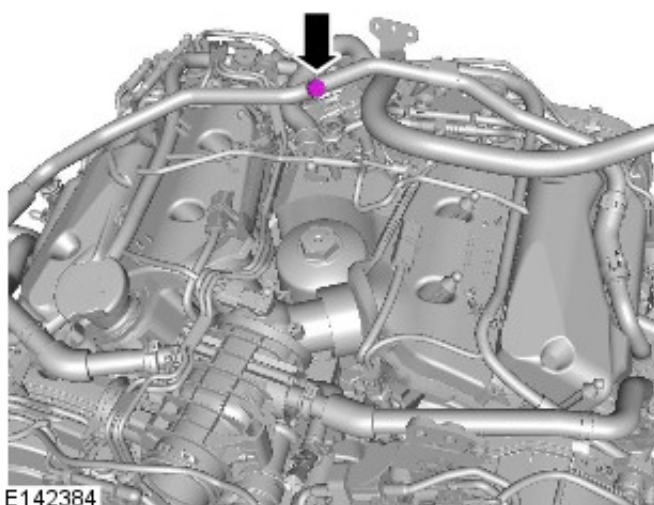
NOTE:

Once coolant is visible from the outlet, close the bleeding points.

26.



27.



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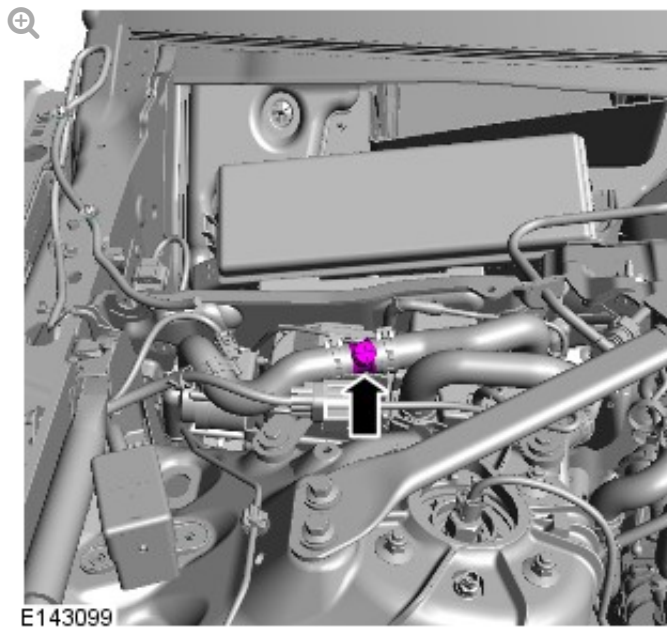
28. Set the heater controls to maximum.

29. Start the engine and allow to idle.

30.

NOTE:

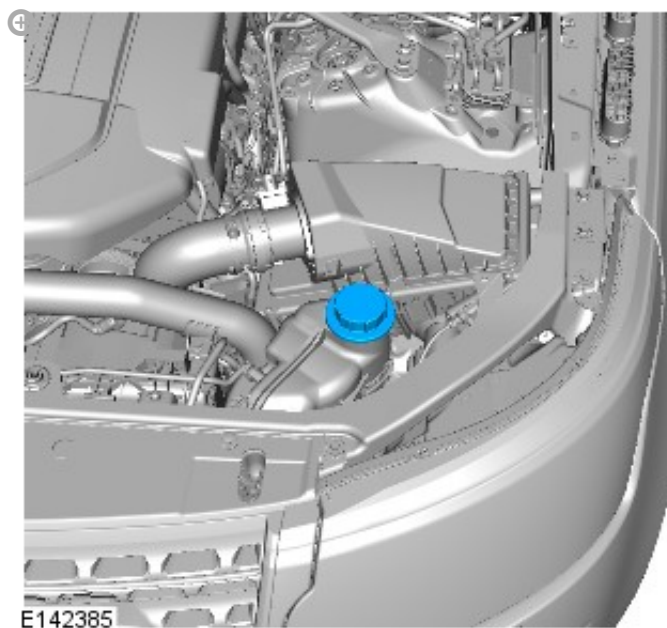
Where fitted.



Once coolant is visible from the outlet, close the bleeding points.

31. Hold the engine speed at 2000 rpm until warm air is expelled from the heater.
32. Continue to fill the coolant expansion tank to the rim until the coolant level stops dropping.
33. Install the coolant expansion tank.

34.



Install the expansion tank cap after making sure that the final coolant level is 40mm above the maximum cold fill line.

35. Run the engine until the thermostat opens and hot air is emitted from the front and rear face vents.

36. Switch the engine off and allow to cool.

37.

WARNINGS:

- When releasing the cooling system pressure, cover the coolant expansion tank cap with a thick cloth.
- Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.

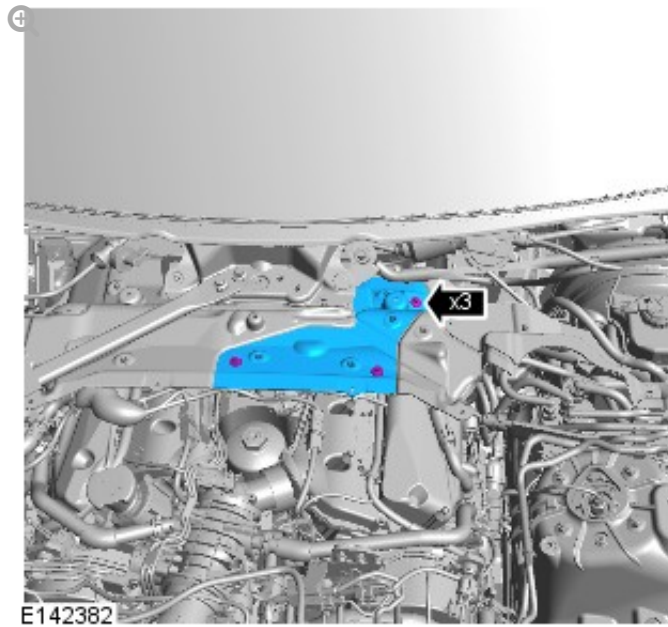
CAUTION:

Make sure the coolant level remains above the "COLD FILL RANGE" lower level mark.

Check and top-up the coolant if required.

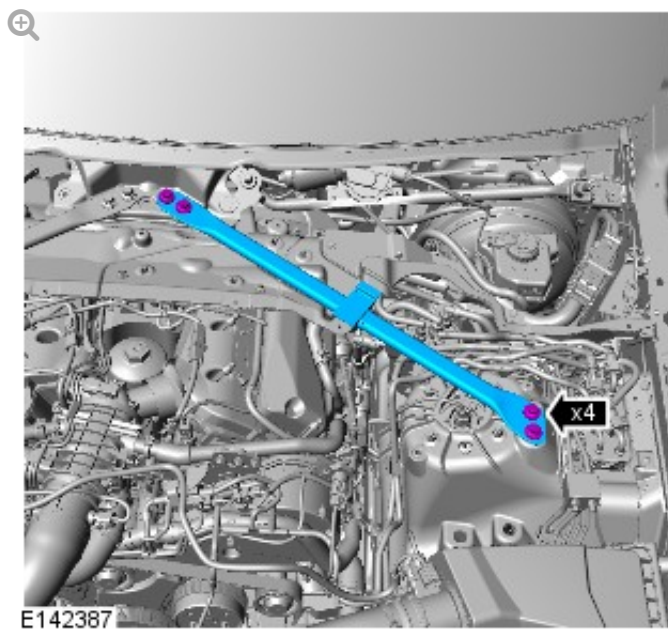
38. Visually check the engine and cooling system for signs of coolant leakage.

39.



Torque: 6 Nm

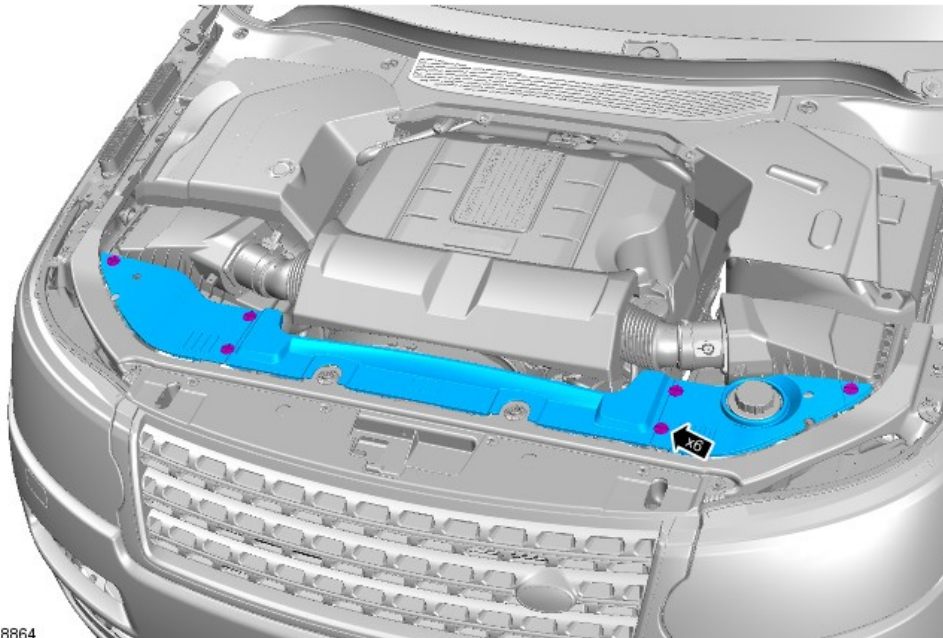
40.



Torque: 55 Nm

41. Refer to: [Cowl Panel Grille](#) (501-02 Front End Body Panels, Removal and Installation).

42.



43. Refer to: [Engine Cover - TDV6 3.0L Diesel](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).
44. Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).

NOTE:

Once coolant is visible from the outlet, close the bleeding points.